Predicting 2025 Strikeout Rate

Phillies Quantitative Analyst Associate (2027) Trial Project: Methodology

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Summary

I predicted 2025 strikeout rate (K%) for all 873 pitchers in `k_2026.csv`. Weighted by batters faced (TBF), the predictions have an RMSE of 0.0411 and an MAE of 0.0317, so a typical miss is about 3.2 K% points, and they explain 41% of the pitcher-to-pitcher variation in 2025 K% (R^2). For comparison, the 2024 league average explains none of that variation and each pitcher's own recent K% explains 9% (Table 1).

Each prediction also comes with an 80% interval, and 80% of actual 2025 values landed inside theirs, at every workload level. For an established pitcher striking out 22% of batters, the interval runs about 4.3 points either side of the prediction over 600 TBF and about 7.8 points either side over 60 TBF.

What drove the predictions was mostly a pitcher being himself: his own strikeout history, stabilized so a small sample isn't taken at face value, followed by last season's Stuff+, whiff rates, and velocity. A large share of the error that remains is sampling noise that no model can remove (Figure 2), and none of the variants I tested, whether new features, different model types, or different shrinkage, moved the backtest RMSE by more than 0.0003 (Table 4). Further accuracy would have to come from information this data doesn't contain.

Table 1: 2025 accuracy, all players, weighted by TBF. Log-loss is per batter faced; lower RMSE/MAE/log-loss and higher R^2 are better.

Method	RMSE	MAE	R^2	Log-loss
League average K% from 2024 (naive)	0.0539	0.0419	0.00	0.5297
Pitcher's own recent K%, league average if none (naive)	0.0512	0.0383	0.09	0.5311
Final model	**0.0411**	**0.0317**	**0.41**	**0.5261**

Results

Table 2: 2025 accuracy by pitcher group (TBF-weighted).

Group	Players	Median TBF	RMSE	MAE	80% interval coverage
Established pitchers	671	195	0.0400	0.0311	79%
Newcomers with minor-league data	88	79	0.0428	0.0316	89%
Newcomers without minor-league data	59	84	0.0606	0.0463	80%
Position players	55	9	0.0484	0.0273	82%

Established pitchers, who faced 90% of all batters, were predicted most accurately, and newcomers without a minor-league record were the hardest group because the least is known about them. For position players, the calibration I fixed from 2022-2024 raised the 2025 RMSE (0.0465 before, 0.0484 after), because 2025 position players struck out more often than recent ones. I kept it anyway, since the rule was set on earlier seasons and changing it after seeing 2025 would stop 2025 from being a true out-of-sample test.

Grouped into deciles by prediction, observed 2025 K% tracks predicted K% with a slope of 1.00, where 1.00 would mean the predictions spread out exactly as much as reality did. The one systematic error came from the league as a whole. League K% fell from 22.6% in 2024 to 22.2% in 2025 while the forecasts averaged 22.7%, and nothing available before Opening Day pointed to a league-wide drop. That gap accounts for only 1% of the squared error, but it does mean part of any individual pitcher's 2025 miss reflects the league environment and not his own skill.

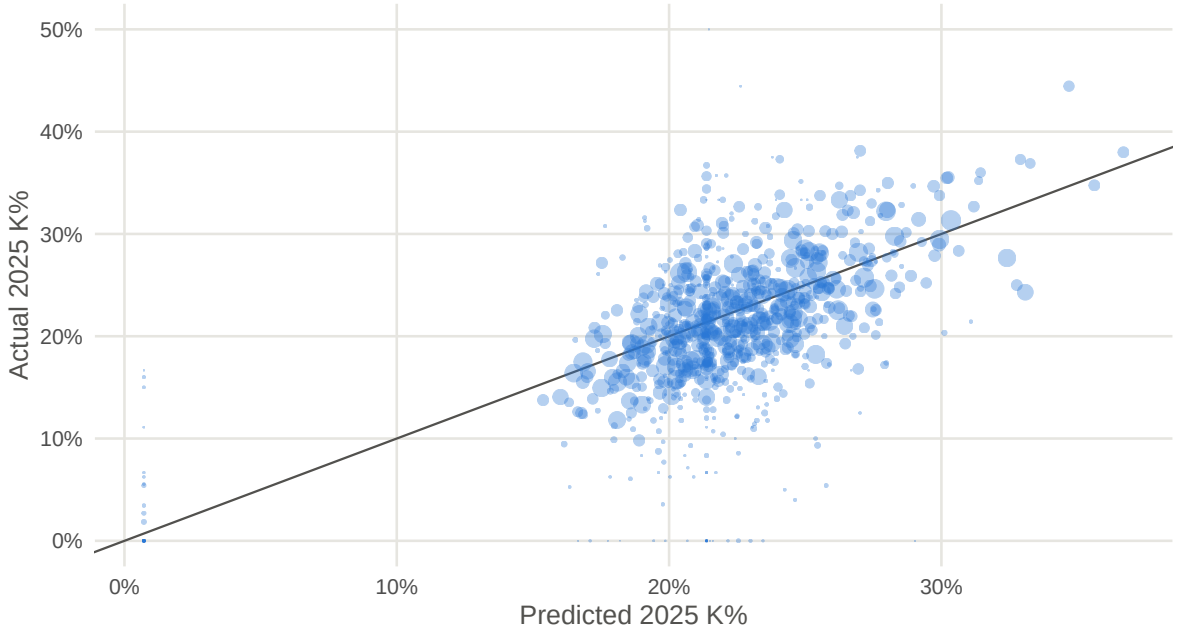


Figure 1: Predicted vs. actual 2025 K%. Point area is proportional to TBF; the line is a perfect prediction. Small-TBF points scatter widely because of sampling noise. The vertical column near 1% is position players.

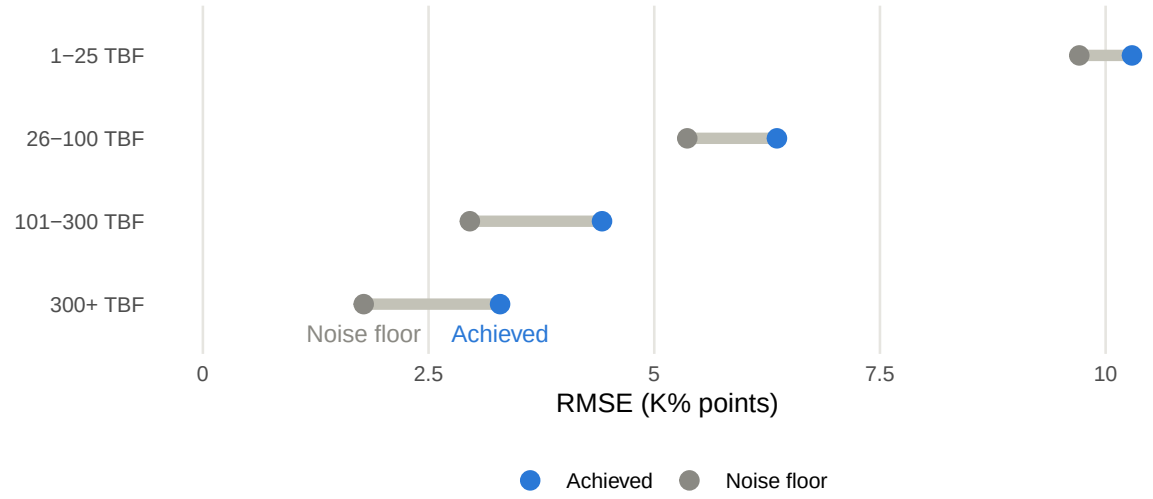


Figure 2: 2025 prediction error by workload (TBF-weighted RMSE). The grey point is the error a perfect estimate of true talent would still make from sampling noise alone; the gap to the blue point is what a model can still improve.

Method

Four pitcher groups, one routing rule

A veteran starter, a debuting rookie, and an infielder mopping up a blowout need different treatment, so I routed each player to one of four groups (Table 3). The routing uses only seasons before the one being predicted, which means each group holds the same kind of player in 2022, 2023, 2024, and 2025, and each group's backtest measures the kind of player it is later applied to.

Table 3: Pitcher groups. The first matching rule wins, in order: position player, established, newcomer.

Group	Who	Prediction rule	2025 players
Established pitchers	Pitched in an earlier season of k_2026.csv	Blend of three models (below)	671
Newcomers with minor-league data	No earlier k_2026.csv season; has a prior MiLB season	Elastic net on prior MiLB stats	88
Newcomers without minor-league data	No earlier k_2026.csv season or MiLB record	Prior season's league-average K%	59
Position players	Has an MLB hitting season ($\geq$25 PA) and no MLB pitching record	Prior season's position-player K%	55

Established pitchers: three models, blended

Every input for these pitchers comes from before the season being predicted: K% from the prior one and two seasons, a TBF-weighted K% over the last three seasons pitched, last season's Stuff+,

whiff%, zone-swing-miss%, chase rate, walk rate, and CSW%, fastball velocity, velocity and spin for six pitch types, role (share of appearances as a starter), age, career TBF, and the prior season's league K%.

I combined three models, each good at something different. Empirical Bayes shrinkage pulls a pitcher's recent K% toward the league average, $\hat{p} = (K + k\,m)/(\text{TBF} + k)$ over the last three seasons, where m is the prior season's league K% and $k = 180$ TBF is fit on the training seasons. That is the beta-binomial posterior mean [1], so a pitcher's own rate earns weight only as fast as his batters faced accumulate. The fitted k is larger than the roughly 70 batters faced at which K% stabilizes within a season [2], as it should be: a forecast of next season has to absorb real changes in talent from one year to the next as well as sampling noise. A binomial elastic net [3], a penalized logistic regression on strikeout counts, uses every input above. XGBoost with a logistic objective [4] adds boosted trees that pick up non-linear effects and interactions, including any between age and Stuff+.

None of the models' settings are hand-picked. Inside every training window, 5-fold cross-validation on binomial log-loss chooses the elastic net's balance between L1 and L2 penalties ($\alpha = 0.25$ in the final fit) and its penalty strength, plus XGBoost's tree depth (2), its minimum leaf size (50 TBF), and its number of trees (372). The folds keep all of a pitcher's seasons together [6], so the tuning never scores a pitcher the model has already seen, and none of it touches the backtest seasons. The learning rate (0.02) and subsampling rates stay at conventional values, since early stopping compensates for them.

Before the elastic net and XGBoost see them, the K% inputs are shrunk the same way as in the first model, so a 3-for-3 cameo enters as roughly league average instead of 100%. All three models maximize the binomial likelihood of the strikeouts, which keeps every prediction strictly between 0 and 1, and the blend weights are non-negative and sum to one (EB 0.25, elastic net 0.60, XGBoost 0.15).

I tested all of this with a rolling-origin backtest [5]: train on 2022 and predict 2023, then train on 2022-2023 and predict 2024, with nothing tuned on the season being predicted. Over those two seasons the blend's RMSE was 0.0395, against 0.0526 for the league average, 0.0486 for trailing K%, and 0.0412 for shrinkage alone. No modeling choice was made on 2025 results.

Newcomers and position players

Newcomers with minor-league data get a binomial elastic net that translates their most recent prior MiLB season into an MLB K%. On its one backtest season (2024, n = 84) it scored an RMSE of 0.0584 against 0.0593 for the league average, a small edge from a single season of 84 pitchers, which makes it the roughest part of the model. Newcomers without MiLB data get the prior season's league average, and position players get the prior season's position-player average.

Calibration and intervals

I multiply each group's predictions by that group's ratio of actual to predicted strikeouts in its 2022-2024 backtest, which is 0.998 for established pitchers and 0.57 for position players. It is a ratio because K% is a rate, and subtracting a fixed 2-point error from a 1% position-player rate would predict 0%.

The 80% interval combines two independent sources of error: the model's own error, measured in the backtest after removing sampling noise (SD 0.029 for established pitchers), and binomial sampling noise, $p(1-p)/\text{TBF}$. The second term is why the interval is wide for a 15-TBF call-up and narrow for a 600-TBF starter. The 2025 TBF sets only the interval's width for the realized sample and never enters the point prediction.

Evidence for each design choice

Every choice in Table 4 was decided on the 2023-2024 backtest or by a structural test, never by its 2025 result.

What drives the predictions

In the final XGBoost model, the three-season K% carries 42% of the total gain, last season's K% 16%, and last season's Stuff+ 7%. Whiff% and zone-swing-miss% add a real amount, and fastball velocity (4%) matters more than any other single velocity or spin measure. CSW% adds only 0.5%, since it is largely redundant with whiff% and Stuff+ once those are in the model.

Safeguards

The only 2025 value the model uses as an input is Age. 2025 TBF, Stuff+, and Team never enter a prediction, and 2025 K% is used only for scoring. The merge script stops if any outside file contains a season after 2024. Every feature for season t comes from seasons before t, a lag is blanked if its source isn't exactly the season it claims to be, and group membership uses only prior seasons. Imputation medians, the shrinkage constant, and all tuning are computed from the training seasons alone and then applied unchanged to the season being predicted.

The scripts stop on duplicate player-seasons (which would leak a season's K% into its own features), on missing or non-positive TBF, and on text in any numeric input column, which they report by column and value. A test script, `tests/test_pipeline.R`, injects edge cases (0-for-5, 0-for-35, 3-for-3, and 5-for-5 histories, impossible telemetry, a pitcher with no tracking data at all, missing Stuff+ or Age, a duplicated row, and a stray text value) and confirms that every prediction stays inside (0, 1), that tiny samples barely move a forecast, that bad inputs are rejected, and that the 2025 predictions are calibrated. Two consecutive runs produce identical predictions.

Limitations and next steps

If I kept building on this, I'd start with the minor-league translation, which only narrowly beats the league average, on a single backtest season. The Statcast minor-league data (apparently Triple-A only) covers just 2022-2024, and 2022 only partly, so the translation trains on a single season. Translation factors fit across more seasons and levels, with pitch-level detail, would be the first upgrade. Pre-season Steamer or ZiPS projections would be the next input to test under the same backtest-only rule, because they carry level-adjusted minor-league translations exactly where this

Table 4: Design decisions and the evidence behind them.

Choice	Alternative tested	Evidence
Elastic net alpha and xgboost depth / leaf size chosen by pitcher-grouped CV inside each training window	Fixed values (alpha 0.5; depth 4, leaves of 50+ TBF)	Backtest blend RMSE 0.03954 vs 0.03953 and log-loss 0.53369 for both: no accuracy cost, and three hand-picked constants become a procedure; the MiLB translation improves from 0.0594 to 0.0584
No explicit Age x Stuff+ term (XGBoost can still find the interaction)	Add a centered Age x prior Stuff+ interaction	Backtest blend RMSE 0.0396 with it vs 0.0395 without; the elastic net got worse (0.0400 vs 0.0398)
Tree count chosen by pitcher-grouped CV inside the training seasons	Tree count chosen on the validation season itself	2024 RMSE 0.0397 vs 0.0396: the validation-season pick looks 0.0001 better only because it is tuned on the season it is scored on
Binomial elastic net + logistic xgboost	Gaussian elastic net + gaussian gbm	2024 RMSE 0.0395 vs 0.0397 (90% bootstrap interval of the difference includes 0); outputs bounded in (0,1) by construction; gbm is no longer maintained
Shrink trailing and lagged K% before the elastic net and xgboost see them	Raw K% inputs	Backtest blend RMSE 0.0395 vs 0.0396; a 3-for-3 prior season moves the forecast to 23.4% instead of 28.4% (tests/test_pipeline.R)
Leave whiff%, Stuff+ and other tracking rates unshrunk	Shrink them toward the median the same way	Backtest blend RMSE 0.0395 vs 0.0398
Existing feature set	Add induced vertical break, arm angle, chase-miss%, SwStr%, CStr%	2024 RMSE 0.0396 with them vs 0.0395 without
Shrink toward the league average	Separate starter / reliever averages	Shrinkage-only RMSE 0.0410 vs 0.0412, but no gain once blended
One set of blend weights	Weights that switch by current-season TBF	Used 2025 workload, which is unknown before the season, and lost on 2025 out of sample
Calibrate each group by a ratio (actual / predicted strikeouts)	Subtract each group's average error	Subtracting a 2.1-point error from a 0.9% position-player rate predicts 0%
Position player = MLB hitting season of 25+ PA	100+ PA	100 PA misroutes 23 real position players who pitched in 2021-2024

model is weakest. I left them out so every forecast here comes from my own model, and because a 2025 file would need a verifiable date from before Opening Day.

The model also can't see opponent quality or park, so a schedule heavy on high-strikeout lineups isn't reflected. Injuries show up only as lost playing time, which makes a partial season lost to injury look identical to a demotion, and a prior-season injured-list flag would capture the well-documented pattern of pitchers returning from injury. A league-wide K% shift like 2025's lands on every pitcher at once, and nothing before Opening Day predicts it. Position players who pitched combined for only about 11 strikeouts a season in 2022-2024, so their group average moves a lot from year to year.

Reproducing this analysis

Run `data_merge.R`, then `k_percentage_model.R`, then `render_methodology.R` to knit this document. Every number here is computed from the model's output files, and `README.md` lists the data sources, filters, and package versions. The code was developed with assistance from Claude (Anthropic), https://claude.ai, as cited in each script.

References

[1] D. Robinson, "Understanding empirical Bayes estimation (using baseball statistics)," Variance Explained, 2015. http://varianceexplained.org/r/empirical_bayes_baseball/

[2] FanGraphs Library, "Sample Size." https://library.fangraphs.com/principles/sample-size/

[3] J. Friedman, T. Hastie, and R. Tibshirani, "Regularization Paths for Generalized Linear Models via Coordinate Descent," *Journal of Statistical Software* 33(1), 2010. https://www.jstatsoft.org/v33/i01/

[4] XGBoost documentation, "XGBoost Parameters." https://xgboost.readthedocs.io/en/stable/parameter.html

[5] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed., section 5.10, "Time series cross-validation." https://otexts.com/fpp3/tscv.html

[6] D. R. Roberts et al., "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography* 40(8), 2017. https://doi.org/10.1111/ecog.02881